# A13 Hungry Chippos

## 8-Bit SAR ADC in GF180mcu

| Team Member | Role | GitHub Username | Affiliation |
|---|---|---|---|
| Bruno | Top-level Integration & Comparator | @brubru6707 | BrownU(2nd Year) |
| Max | DAC / Cap Array | @maxwell | BrownU (2nd Year) |
| Sam | SAR Logic | @sam581 | BrownU (2nd Year) |
| Emily | Bootstrap Switch for Comparator/SAR | n/a | BrownU (2nd Year) |
| Mimi | 3-Way Switch (Submodule for DAC | n/a | BrownU (2nd Year) |
| Luc | Alternative Comparator Design | n/a | BrownU (2nd Year) |

Schematic Review

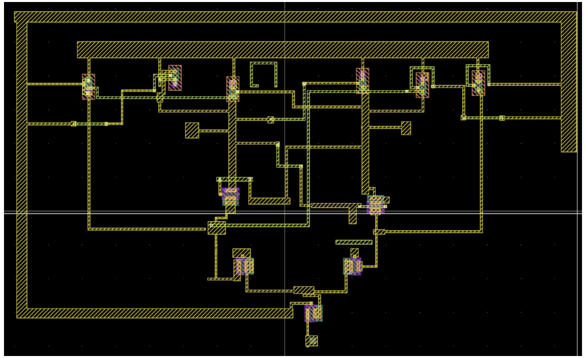
Video | Progress | GitHub Issue



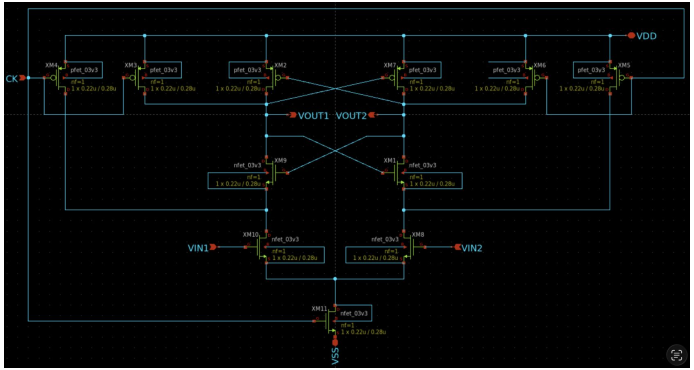
IEEE

# Comparator—StrongARM: General Overview

- Core Architecture: Features a modified StrongARM latch based on the schematic from Behzad Razavi's paper: [The StrongARM Latch].
- Design Motivation: Chosen for its simple design and stability over time compared to alternative architectures.
- Current Performance: Initial Monte Carlo simulations indicate a 30mV offset for the latch.
- Optimization Plan: Following physical layout, a 4-finger design will be implemented on the input transistors to average out voltage mismatches and minimize the systematic offset. The expected amount of mV offset is estimated to be ~10mV reduction
- [Comparator Link]



Comparator Layout (LVS/DRC Checked Off)




Comparator Schematic ^

# Comparator—StrongARM: Other Info

Objectives for the Comparator:

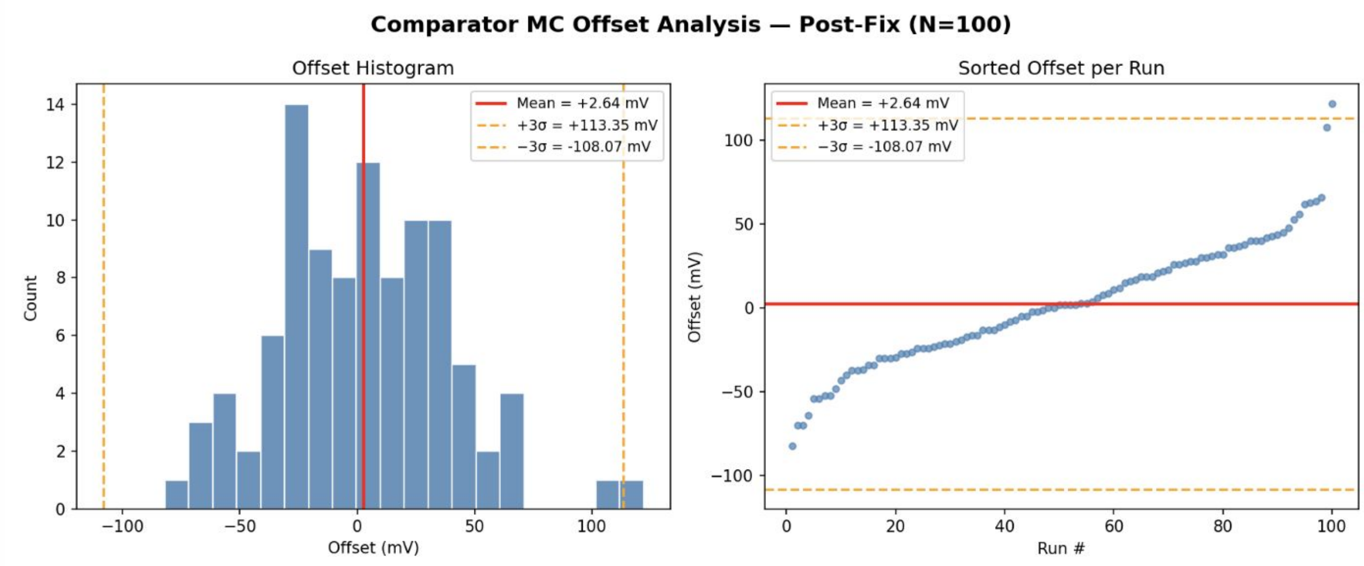
| Task | Goal | Status |
|---|---|---|
| COMP-1/2 | StrongARM latch schematic + symbol | 🟢 Done |
| COMP-3/4 | Functional transient TB + Monte Carlo TB (N≥100) | 🟢 Done |
| COMP-5 (**Gate 1**) | Characterize input-referred offset (MC) + regeneration delay < 2 ns | 🟡 In progress — offset done (mean +2.6 mV, σ=36.9 mV, cleared since offset is a calibratable DC shift for a SAR, no upsizing needed); **delay < 2 ns still unmeasured** |
| COMP-6 | Physical layout in KLayout | 🟢 Done |
| COMP-7 | Sub-block DRC clean | 🟢 Done (density flags deferred to chip-level fill) |
| COMP-8 | Sub-block LVS clean | 🟢 Done (11/11 devices match) |
| COMP-9 | Post-layout extraction (PEX) corner sim | ⚪ Not started |

**What's left to close out the comparator:** measure regeneration delay (<2 ns) to close Gate 1, then run PEX corner simulation (COMP-9). Everything else (schematic, layout, DRC, LVS) is complete.

Note: Got an issue with LVS/DRC for comparator physical layout (bug in the PDK side) but physical layout still works when ran through the terminal

# Comparator—StrongARM: Monte Carlo Loop (100 Sample Run)

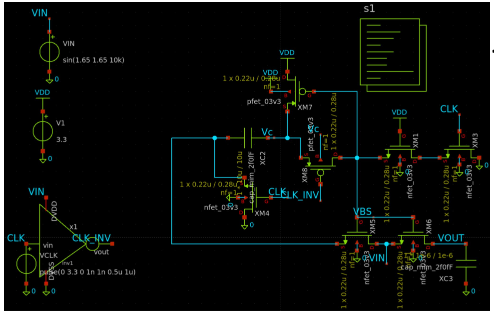
Monte Carlo Comparator Offset (100 runs)
Mean : +2.640 mV
Std : 36.904 mV
3sigma: [-108.071, +113.350] mV
Min : -82.000 mV
Max : +122.000 mV



# Switch: General Overview

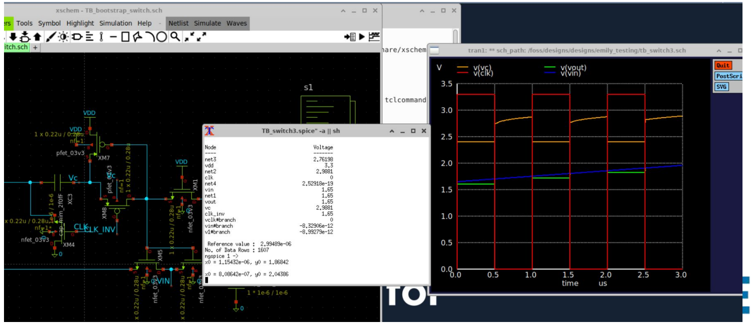
- Architecture & Purpose: Implemented a bootstrapped switch for the sample-and-hold circuit to eliminate input-dependent on-resistance.
- Circuit Operation:
  - Off-Phase: Pre-charges a capacitor to V_DD.
  - Sample Phase: Stacks the pre-charged capacitor on top of the input signal V_IN.
  - Gate Drive: Actively drives the gate of the main sampling transistor to V_DD + V_IN, maintaining a perfectly constant V_GS of 3.3V.
- Simulation & Verification: Isolated and verified the core switching logic by simulating the circuit with an ideal inverter for the inverted clock.
  - Transient plots confirm that the output tracks the input with zero distortion.
- [Schematic Location]

Analog Input → [Bootstrap Switch / S&H] → [Cap Array / DAC] → [Comparator] → [SAR Logic] → Digital Output



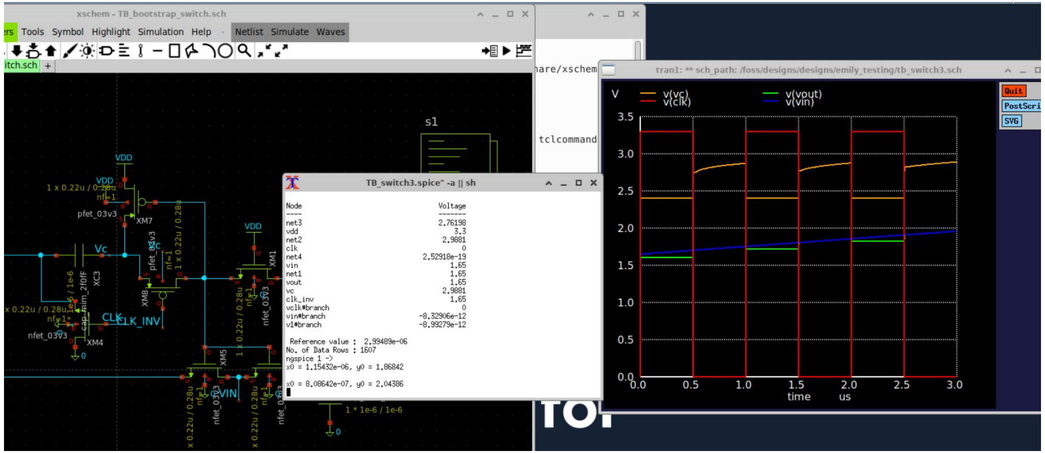

< Bootstrap Switch Schematic

Diagnosis On The Next Slide >



# Switch: TestBench

```
C {code.sym} 390 -310 0 0 {name=s1 only_toplevel=false value="
.param mim_corner_1p0fF=1 mim_corner_1p5fF=1 mim_corner_2p0fF=1
.param mc_c_cox_1p0fF=0 mc_c_cox_1p5fF=0 mc_c_cox_2p0fF=0
.param var_vth=0 var_k=0
.lib /foss/pdks/gf180mcuD/libs.tech/ngspice/sm141064.ngspice typical
.lib /foss/pdks/gf180mcuD/libs.tech/ngspice/sm141064.ngspice cap_mim
.include /foss/pdks/gf180mcuD/libs.tech/ngspice/design.ngspice
.options savecurrents
* Simple Ideal Inverter to generate CLKB from CLK for PMOS M8
Ainv1 CLK CLKB ideal_inv
.model ideal_inv d_inverter(rise_delay=1n fall_delay=1n)
.control
save all
tran 2n 3u
plot v(CLK) v(VIN) v(Vc) v(VOUT)
plot V(vbs)-V(vin)
.endc
"}
```




The code: runs a 3 μs transient simulation and plots the clock, input, bootstrapped node Vc, output, and the bulk-source voltage to check switch behavior and voltage headroom

**-CLK** (red): 0–3.3V clock toggling every 0.5 μs, gating the switch on/off.
-**Vc** (orange): the bootstrapped gate node. It rises above the input each time CLK goes high, sitting around 2.8–3.3V, i.e. boosted above VIN so the switch stays fully on with constant overdrive (the whole point of bootstrapping).
**-VOUT** (green) tracks **VIN** (blue) almost exactly as both ramp from ~1.6V to ~2.0V, meaning the switch passes the input through with negligible voltage error/droop.

The bootstrap circuit is successfully holding a constant gate overdrive each cycle, and the output follows the input cleanly..

# SAR: General Overview

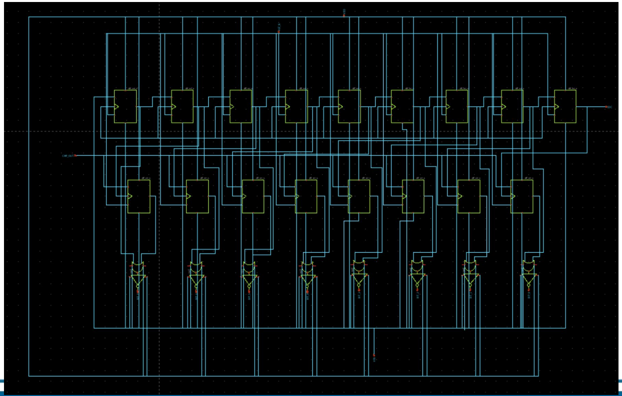
- Architecture & Components: Implemented a custom 17-stage Anderson architecture utilizing 17 Master-Slave flip-flops built entirely from custom CMOS logic cells.
- Functional Layout:
    - Top Row (Sequencer): Passes a logic high pointer bit-by-bit to track the conversion cycle.
    - Middle Row (Code Register): Captures and locks in the comparator's decisions.
    - Bottom Row (OR Gates): Combines signals to drive the capacitor DAC switches.
- Initialization Control: Designed custom active-low asynchronous reset and preset flip-flops to manage the initialization state.
- Current Status & Debugging: Currently resolving an active-low reset handling issue within the NOR gate feedback loop of the preset flip-flop.
    - Finalizing the logic inversion on these NOR gates, after which the fully custom, transistor-level digital controller will be ready for integration and fully operational.
- [SAR Location]

SAR Top
Level Integration >

Sequencer >

Code Register >

OR Gates >

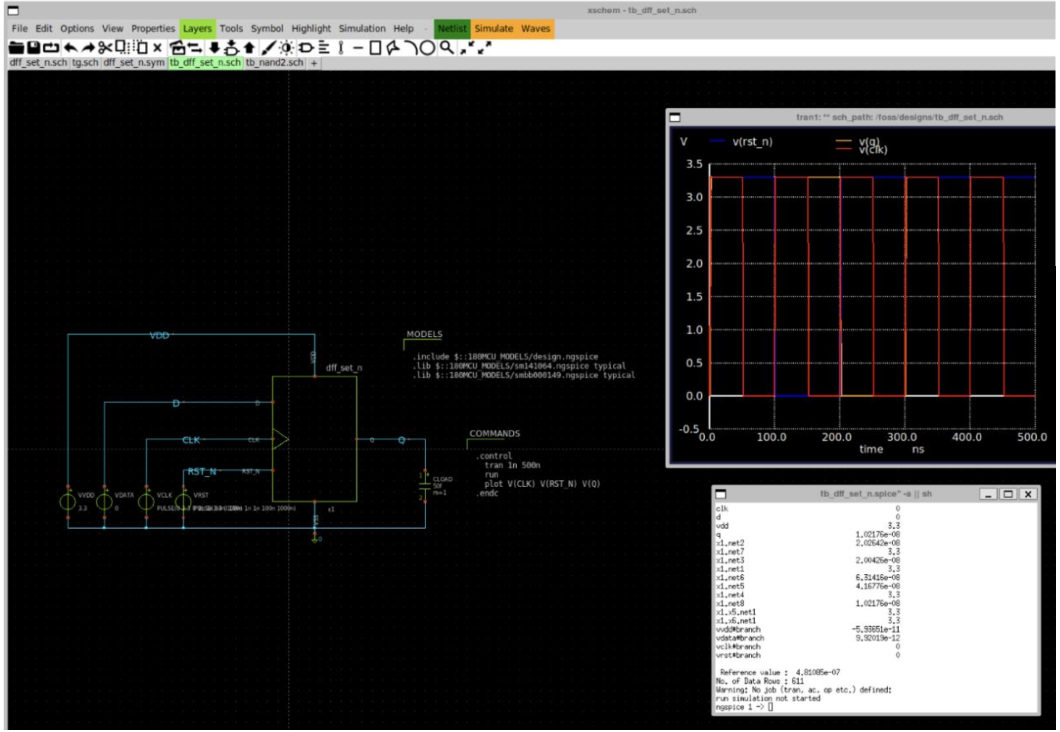


# SAR—Objectives

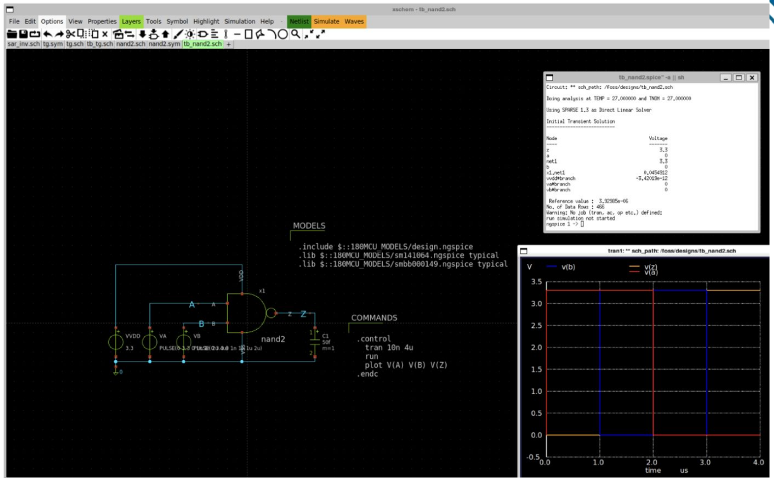
| Task | Goal | Status |
|---|---|---|
| SAR-1 | Standard-cell schematic + testbench: INV, TG, NAND2, NOR2, DFF_RST_N, DFF_SET_N | 🟡 In Progress — 5/6 verified; **DFF_SET_N broken** (NOR-gate/inverted-RST_N bug) |
| SAR-2 | Top-level SAR_LOGIC schematic (17 FFs: 9-FF sequencer + 8-FF code register + 8 OR-gate DAC drive) | 🟡 In Progress — schematic assembled, not yet simulated (blocked on SAR-1 fix) |
| SAR-3 | Full binary-search functional testbench (verify code capture + EOC) | ⚪ Not Started |
| SAR-4 | SAR_LOGIC physical layout (KLayout) | ⚪ Not Started |
| SAR-5 | Sub-block DRC clean | ⚪ Not Started |
| SAR-6 | Sub-block LVS clean | ⚪ Not Started |

# SAR: Submodules Schematics + TestBenches

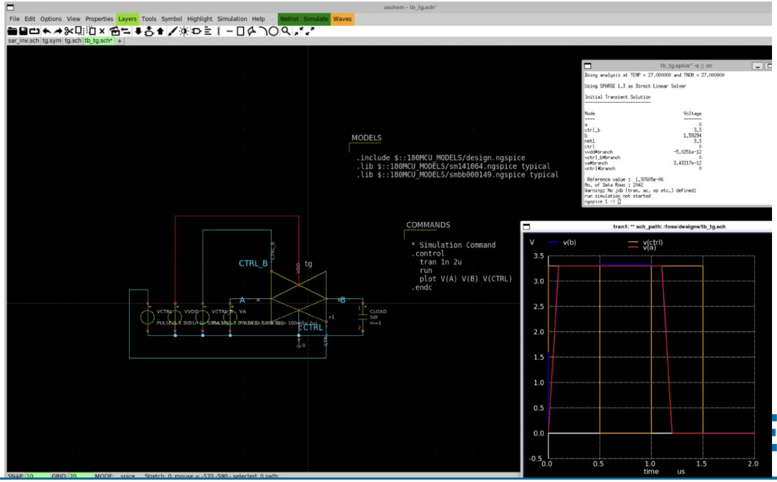
## DFF with Active-Low Asynchronous Preset




## NAND2



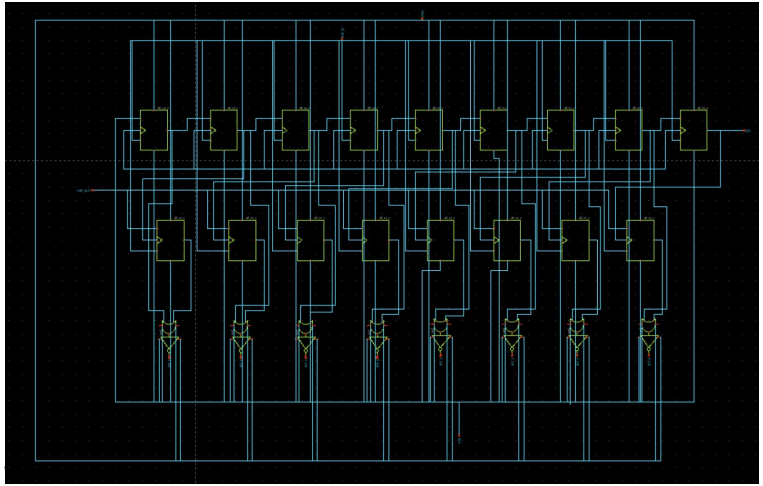

## Transmission Gate



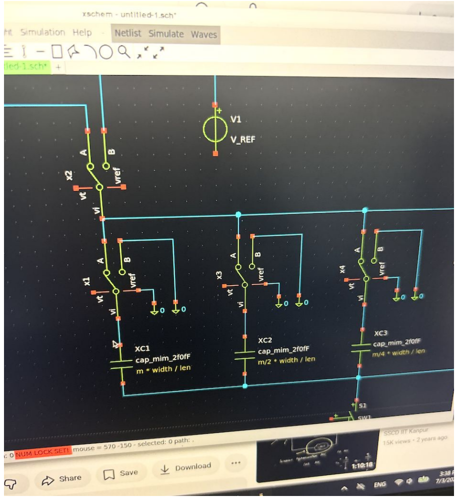
# DAC

We want to showcase the top-level integration schematic of the SAR logic again.  The bottom module is where the CDAC integrates. While the final 8-bit binary-weighted MIM capacitor array is in progress due, we have completed and verified the 3-way NMOS switch submodule that operates alongside it.

We tested our design by putting in fake, temporary capacitors to see if the electronics worked. The test proved that our switches correctly grab the input signal during sampling, and smoothly flip to the reference voltage or ground during the guessing phase.



3-Way NMOS Switch Schematic  >



# DAC—Objectives

| Task | Goal | Status |
|---|---|---|
| DAC-1 | Binary-weighted 8-bit cap array schematic (C_u ≥ 50 fF) — now also needs to plan for 8 single-ended digital switch-control pins from the SAR controller | ⚪ Not Started |
| DAC-2 | Cap array symbol | ⚪ Not Started |
| DAC-3 | 256×C_u switching & settling time testbench | ⚪ Not Started |
| DAC-4 | **Gate 2** — 0.5 LSB settling within 40 ns (TT) | ⚪ Not Started |
| DAC-5 | Unit-cell layout, common-centroid placement (KLayout) | ⚪ Not Started |
| DAC-6 | Full array layout with dummy/fringe peripheral caps | ⚪ Not Started |
| DAC-7 | Sub-block DRC clean | ⚪ Not Started |
| DAC-8 | Sub-block LVS clean (Netgen) | ⚪ Not Started |

# References

Comparator
https://www.seas.ucla.edu/brweb/papers/Journals/BR_Magzine4.pdf
https://drive.google.com/file/d/1b4CQ6VKFIqiCHWUxqPrQSjMsSP0B4uAp/view?usp=sharing
https://github.com/iic-jku/IIC-OSIC-TOOLS/issues/307

Switch
https://ieeexplore.ieee.org/document/9531736 - 5) Lin-Hsieh Bootstrapped Switch
https://www.researchgate.net/publication/2983201_Switch_Bootstrapping_for_Precise_Sampling_Beyond_Supply_Voltage

DAC
xxxx

SAR
https://notebooklm.google.com/notebook/a845c4f2-c32e-43c8-9e55-b57e1f3be735?authuser=2
Analog Guide:
https://github.com/sscs-ose/sscs-chipathon-2026/blob/main/resources/Analog/eda/README.md
Local Repo: https://github.com/SamS581/hungry-chippos-chipathon-2026

Information
https://en.wikipedia.org/wiki/Successive-approximation_ADC
- Cycles through all bits by turning the nth bit into a 1, resetting to 0 if $V_{in} < V_{cmp}$ else keeping

https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10704295
- Optimize speed, resolution, and power

https://techschems.com/sar-adc-schematic
https://pages.jh.edu/aandreo1/216/Archives/2014/Handouts/POP_Ch3-4.pdf
- Logic gates from NFETs and PFETs + Rules

Example Designs
https://www.yilectronics.com/Students/tsnakai/Labs/Project/Project.html
- Make NAND → make D flip flop → 8-bit SAR
- Tested at each step

https://www.ee.columbia.edu/~kinget/EE6350_S16/03_ADC_Dhruv_Gurkaran_Shikhar/schematic.html
- 16 flip flops (8 shift, 8 control)

https://gemini.google.com/share/b2b6cdf181c7
https://share.google/aimode/s1aaorTjUL1QYcPYM
- Option 1: xschem with multiple shift register+ring counters
- Option 2: Verilog with coded block

https://www.yilectronics.com/Courses/ENGR338_CE/f2021/proj/SARADC_LTSpiceReport.pdf